

Foreign Object Debris Detection Assessment

YOLO baseline evaluation and deployment proposal

5-epoch checkpoint | Held-out test evaluation | Confidence threshold 0.25

Executive summary

This report documents a FOD object detector trained on the cleaned FOD dataset and evaluated on a held-out test set. The preserved 5-epoch checkpoint achieved 93.6% precision, 95.1% recall, 97.6% mAP@50, and 72.1% mAP@50-95 on 5,069 test images. The results are promising as a baseline, but small-object localization remains the main limitation and the model should not be treated as safety-certified.

Because of the available time, the experiment used early stopping and the 5-epoch checkpoint was the reliable completed result. With more time, the next experiment should use approximately five times more diverse data, including more small debris, altitudes, lighting, weather, and hard negatives. This is expected to improve robustness, especially for small and low-contrast objects, but it must be verified by a new held-out test set.

1 Dataset and exploratory analysis

The project uses a cleaned YOLO-format FOD dataset derived from the public FOD image collection. It contains 31 object classes representing common runway debris such as bolts, nuts, wires, tools, luggage parts, plastic parts, paint chips, rocks, and metal components. The cleaned split contains 23,655 training images, 5,069 validation images, and a separate test split used for final evaluation. Images were standardized to 640 x 640 during training and inference.

The classes are imbalanced. Common categories such as Bolt, Pliers, Wrench, Wire, and PlasticPart have more examples than rare categories such as Tape, Screw, Wood, and Hose. Challenging cases include small objects, pavement-like texture, occlusion, and changing illumination.

Split	Images	Purpose
Train	23,655	Model fitting and augmentation
Validation	5,069	Training-time model selection
Test	5,069	Held-out final evaluation

2 Training configuration

Detector: YOLOv8s. Image size: 640 x 640. Batch size: 16. Completed epochs: 5. Early stopping patience: 5. Optimizer: AdamW. Augmentations: rotation, translation, scale, horizontal flip, and HSV changes. Inference threshold: 0.25.

The model was evaluated using the best weights from the completed 5-epoch run. A 15-epoch run was attempted separately but was not treated as valid because slow image loading made it impractical to complete reliably within the available time.

3 Held-out test results

Test images: 5,069. Precision: 0.936. Recall: 0.951. mAP@50: 0.976. mAP@50-95: 0.721. AP@75: 0.885. AP small: 0.604. AP medium: 0.747. AP large: 0.762.

The high mAP@50 shows that the detector generally identifies the correct object at the standard 0.50 IoU threshold. The lower mAP@50-95 and small-object AP show that localization is less consistent at stricter overlap requirements and for small debris.

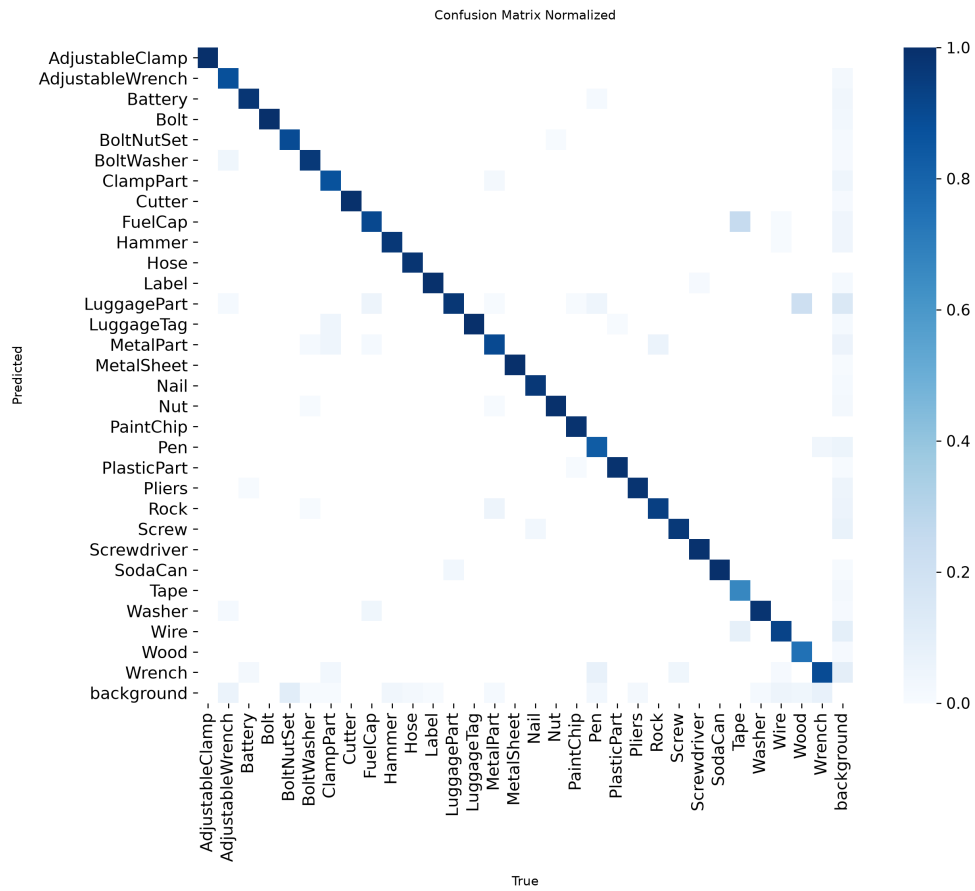


Figure 1. Normalized confusion matrix from the held-out test split.

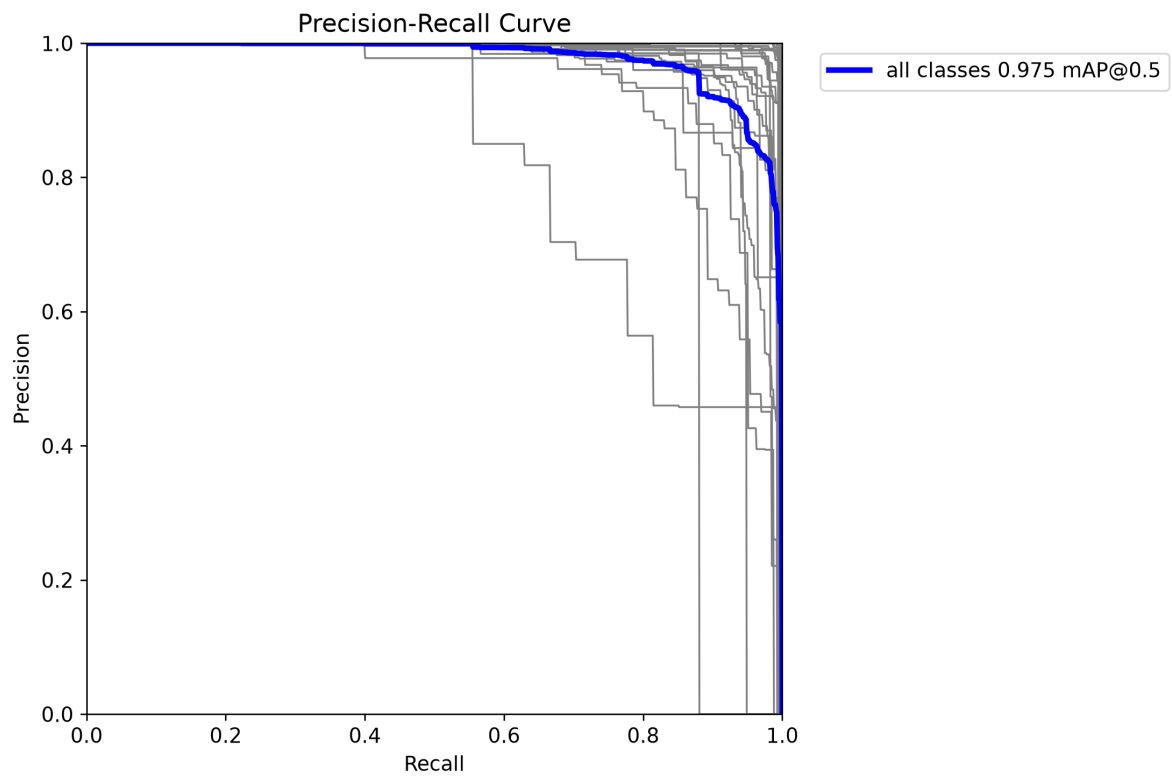


Figure 2. Precision-recall curve.

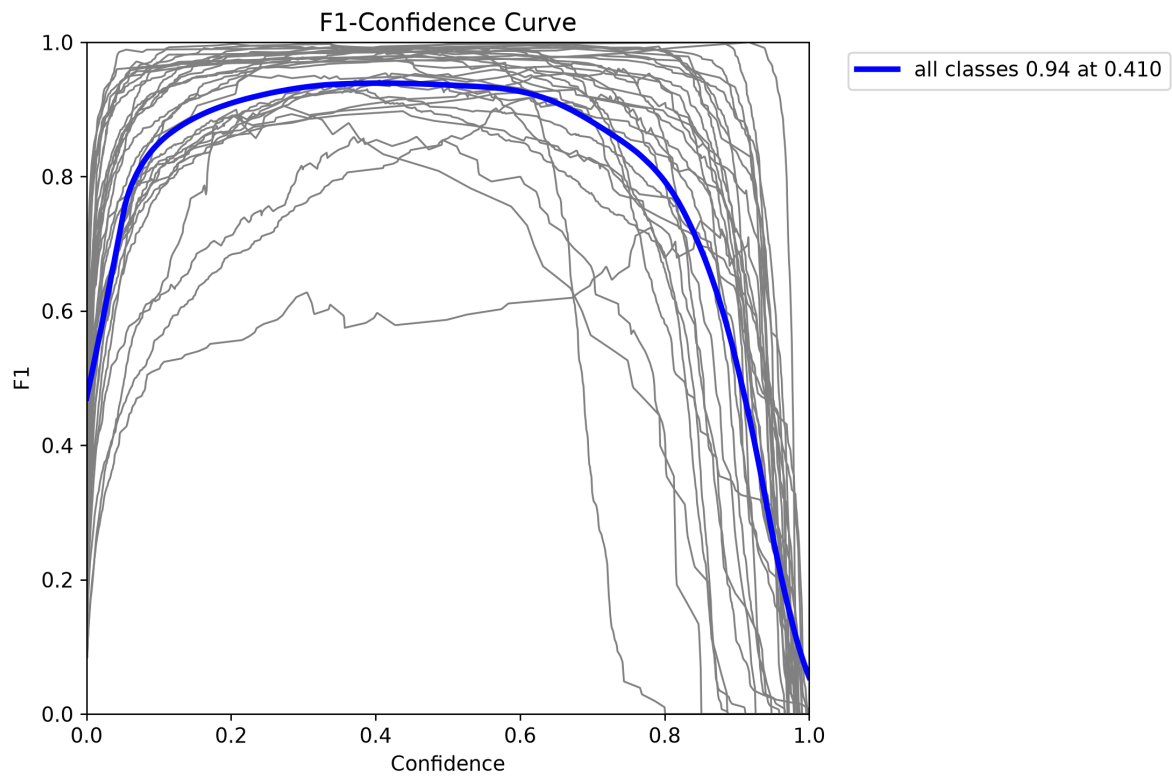


Figure 3. F1 curve across confidence thresholds.

4 Inference examples and interpretation

Inference was run on all 5,069 test images with confidence threshold 0.25. Annotated images and YOLO label files were saved for review. The label files contain class IDs, normalized box coordinates, and confidence scores.



Figure 4. Example test inference.



Figure 5. Example held-out inference.



Figure 6. Additional annotated inference example.



Figure 7. Validation batch predictions compared with labelled objects.

5 Error analysis

False positives can occur when shadows, runway markings, reflections, cracks, stains, or textured pavement resemble a debris class. Similar-looking tools and metal parts can also be confused when objects are small or partially visible.

False negatives are most likely for small objects, low-contrast objects, heavy occlusion, unusual viewing angles, and rare classes with limited examples. The small-object AP of 0.604 is evidence that this regime needs additional work.

For runway safety, a missed object is generally more serious than a false alarm. The initial threshold of 0.25 therefore favours recall. In deployment, the threshold should be selected from the validation precision-recall curve against an explicit false-alarm budget, followed by human review, temporal tracking, or a second-stage verification model.

Concrete improvements are: collect at least five times more varied data; use tiled or higher-resolution inference for small debris; and mine hard negatives from runway shadows, markings, reflections, and texture while balancing rare classes.

6 Edge AI deployment approach

A practical UAV deployment would use a stabilized RGB camera connected to an NVIDIA Jetson Orin Nano or Jetson Orin NX. Frames would be resized or tiled on the Jetson, processed by a TensorRT-optimized YOLO model, and filtered by confidence and persistence across consecutive frames. GPS and IMU data would attach a location and timestamp to each alert. The operator dashboard would receive the object class, confidence, crop, coordinates, and a short event clip.

The system should fail safely: uncertain detections should be flagged for review; low visibility, GPS loss, camera blockage, and low battery should be reported; and model drift should be monitored across airports and lighting conditions. The detector is an inspection aid, not a replacement for certified airport safety procedures.

7 Further experiments for false-positive reduction

The most defensible next experiment for Task 4 is a two-stage FP-rejection classifier cascade. YOLO proposes boxes; a small CNN classifies each crop as real debris or false-positive such as a shadow, reflection, or runway marking. This decouples false-positive rejection from YOLO training and can reduce the false-alarm rate without changing the detector weights.

Two Stage False Positive Rejection Cascade

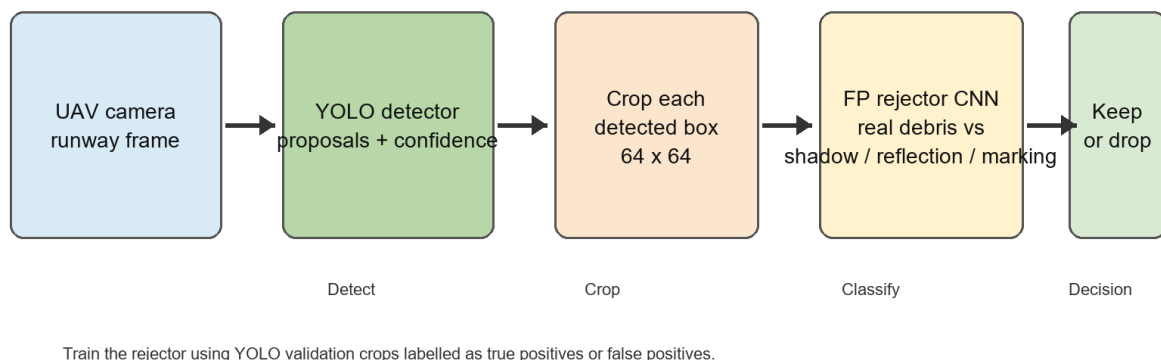


Figure 8. Two-stage cascade for rejecting shadow, reflection, and marking false positives.

Pipeline: run YOLO inference, crop each detected box and resize it to approximately 64 x 64, pass the crop through FPRejectorCNN, and drop detections classified as false positives. Train the classifier on crops labelled from YOLO true-positive and false-positive outputs on the validation set.

Additional experiments should include shadow and reflection hard-negative mining, day/night and fog augmentation, altitude and zoom variation, tiled inference for small debris, and threshold sweeps that report the recall at a fixed false-alarm budget.

8 Reproducibility and files

Model checkpoint: `fod_detection_assessment/runs/fod_yolov8s_gpu_5ep/weights/best.pt`

Metrics and plots: fod_detection_assessment/runs/test_metrics_5ep_final/

Annotated predictions: fod_detection_assessment/submission/test_predictions_5ep_final/

Prediction JSON: fod_detection_assessment/runs/test_metrics_5ep_final/predictions.json